4 Two small speakers, L_1 and L_2, are separated by 15 cm, as shown in Fig. 4.1. A microphone M is moved along line RS parallel to the line joining the two speakers and at a perpendicular distance of 1.5 m away. P is a position on the line RS and equidistant from both speakers.

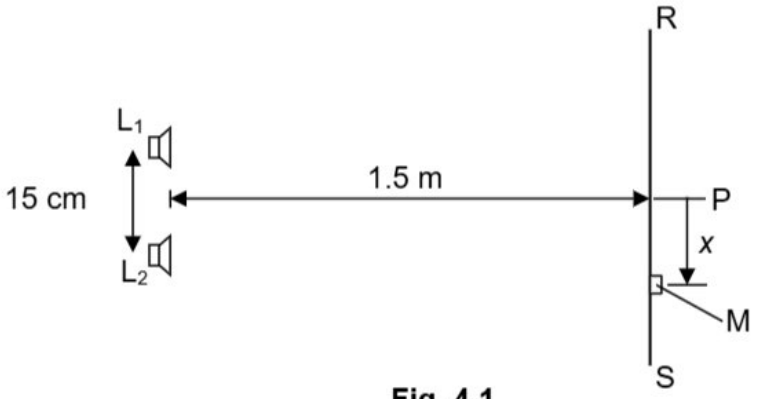


Fig. 4.1

The sound intensity detected by the microphone M varies with displacement *x* as M is moved from R towards S as shown in Fig. 4.2.

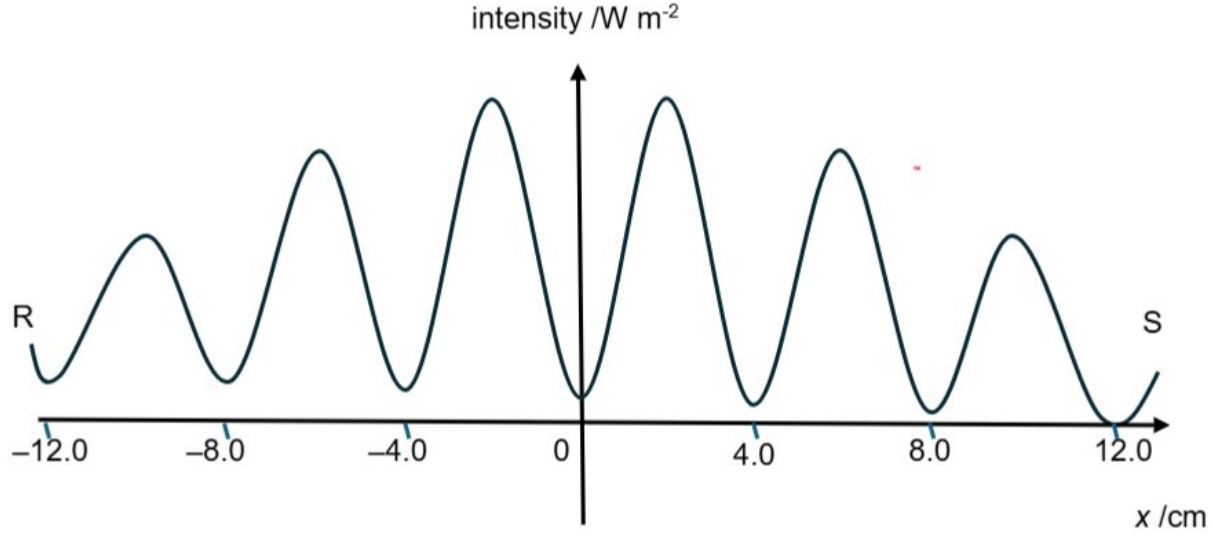


Fig. 4.2

(a) Explain how it can be deduced that the waves emitted by the two speakers are coherent.

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..... [1]

(b) Explain, with reference to Fig. 4.2, how it can be deduced that the sound waves emitted by the speakers are not in phase.

.....

.....

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.....

.....

..... [3]

(c) The speed of sound is 340 m s ^{1}.

Calculate the frequency of the sound waves emitted by the speakers.

frequency = ... Hz [2]

(d) Explain which speaker has a higher power.

.....

.....

.....

..... [2]

[Total: 8]